MILITARY SPECIFICATION

ELECTRONICS - MCD ASSEMBLY FOR MINE, ANTI-TANK: BLU 91/B

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains quality assurance provisions for the fabrication and inspection of parts, assembly and packing of an electronic assembly for Mine, Anti-Tank: BLU 91/B.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging

MIL-P-14232 - Parts Equipment and Tools for Army Materiel, Packaging of

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-E-17555	-	Electronic and Electrical Equipment Accessories, and Repair Parts, Packaging and Packing of
MIL-S-19500/127	-	Semi-conductor Device, Diode Silicon, Voltage-Regulator Types IN4370A through IN4372A, IN746A through IN759A, IN4307A-1 through IN4372A-1, IN746A-1 through IN759A-1, JAN, JANTX, JANTXV, JANS
MIL-S-19500/240	-	Semiconductor Device, Diode, Silicon, Rectifier, Types IN645, IN647, IN649, IN645-1, IN647-1, IN649-1, TX and Non-TX Types
MIL-S-19500/255	-	Semiconductor Device, Transistor, NPN, Silicon, Switching Types 2N2221, 2N2221A, 2N2222, and 2N2222A JAN, JANTX, JANTXV, and JANS
MIL-S-19500/291	-	Semiconductor Device, Transistor, PNP, Silicon Switching Types 2N2906, 2N2906A, 2N2907 and 2N2907A JAN, JANTX, JANTXV, and JANS
MIL-S-19500/406	-	Semiconductor Device, Diode Silicon, Voltage Regulator Types JAN and JANTX IN4460 thru IN4496
MIL-R-39008/4	-	Resistor, Fixed, Composition (Insulated) Established Reliability Style RCR05
MIL-C-39014/5	-	Capacitor, Fixed, Ceramic Dielectric (General Purpose), Established Reliability, Style CKR11
MIL-S-45743	-	Soldering, Manual Type, High Reliabiity, Electrical and Electronic Equipment
MIL-S-46844	-	Solder Bath Soldering of Printed Wiring Assemblies

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MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-202 - Test Methods For Electronic and Electrical Component Parts

MIL-STD-275 - Printed Wiring for Electronic Equipment

MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for

MIL-STD-750 - Test Methods for Semi-conductor Devices

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.11)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

9330379 - Electronics - MCD Assembly
9349483 - Mine Assembly, BLU 91/B, Test, AT40

INSPECTION EQUIPMENT DRAWINGS

1183AS616 - Standard Wave Forms

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

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(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained for the contracting activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the appplicable drawings, specification and standards.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on (Dwg) 9330379, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Physical characteristics.

3.3.1 Envelope and dimensions. The envelope and dimensions of the Electronics - MCD Assembly shall conform to the requirements on Dwg. 9330379, as applicable.

3.3.2 Potting material. Potting material shall comply with the requirements of drawings (Dwgs.) 9287115 and 9287148, and shall be manufactured and dispensed in accordance with the processes, procedures, and controls delineated therein . The potting material shall be tested as specified in 4.4.3.1.

3.3.3 Potted electronic assembly. The potted Electronics Assembly shall comply with Dwg. 9330378 and as specified in 4.5.2.

3.4 Enviromental conditions.

3.4.1 Transportation vibration. The Electronics - MCD Assembly shall be safe to handle and transport and shall show no evidence of internal or external damage that will effect the functions following transportation vibration Test No. 119, Procedure 2 of MIL-STD-331.

3.4.2 Temperature conditioning. The Electronics-MDC Assemblies shall function as specified in 3.5 after conditioning at any of the following temperatures:

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a. Cold, minus 40° - 5°F for four (4) hours minimum (min.).

b. Ambient, for four (4) hours min.

c. Hot, 145° + 5°F for four (4) hours min.

3.5 Functioning. After being conditioned as specified in 3.4, the Electronics-MCD Assemblies, assembled in a test mine, shall be capable of arming and functioning high order after being subjected to high velocity impact, by sensor or self-destruct (SD) modes.

3.6 Electronic Assemblies performance characteristics.

3.6.1 Quiescent current. The Electronics-MCD Assembly quiescent current shall be less than 288 microamperes when the supply voltage is 6.05 ± 0.15 volts at ambient temperature and the input/output terminals loaded as specified in 4.5.5.1.

3.6.2 Primary Time Base (PTB) oscillator period. The PTB oscillator period shall be verified when tested as specified in 4.5.5.2.

3.6.3 Test Time Base (TTB) oscillator period. The TTB oscillator period shall be verified when tested as specified in 4.5.5.3.

3.6.4 S&A assembly arm signal. The PTB clock count at which the S&A arm signal functions the piston actuator firing circuit shall be verified when tested as specified in 4.5.5.4.

3.6.4.1 S&A arm signal inhibit. The S&A arm signal shall not occur when the TTB oscillator is inhibited. This function shall be verified when tested as specified in 4.5.5.4.1.

3.6.5 Electronic arm. The electronic assembly shall not process destruct signals from the low voltage detect, or prime target sensor circuits until after electronic arm. The PTB clock count that initiates electronic arm and the ability of the electronic assembly to inhibit destruct signals prior to electronic arm shall be tested for as specified in 4.5.5.5.

3.6.6 Low voltage detect. The supply voltage level at which the low voltage detect circuit functions the detonator firing circuit shall be verified when tested as specified in 4.5.5.6.

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3.6.7 SD1 self-destruct signal The PTB clock count at which the SD1 self-destruct signal functions the detonator firing circuit shall be verified when tested as specified in 4.5.5.9.

3.6.8 SD2 self-destruct signal. The PTB clock count at which the SD2 self-destruct signal functions the detonator firing circuit shall be verified when tested as specified in 4.5.5.8.

3.6.9 SD3 self-destruct signal. The PTB clock count at which the SD3 self-destruct signal functions the detonator firing circuit shall be verified when tested as specified in 4.5.5.7.

3.6.10 Target sensor functioning. The Electronics Assembly shall provide a fire signal (3.6.10.2) when subjected to valid target signatures. The specific conditions and signal limit under which this function shall occur are specified in Drawing 9332388, and the following paragraphs.

3.6.10.1 Spurious output signal. There shall be no output occurring at the Target Sensor output (Test Point 1) for a minimum of 10 seconds prior to electronic arming after initial application of V_B when tested as specified in 4.5.5.10.

3.6.10.2 Sensor output voltage. The output shall be a minimum of 4 volts when the assembly is subjected to a valid target signature as specified in 4.5.5.10.

3.6.11 Primary Time Base oscillator (PTB) check. The oscillator check circuit shall initiate the function of the detonator firing circuit when the primary time base oscillator period becomes greater than the test time base oscillator period. This requirement shall be tested for as specified in 4.5.5.11.

3.6.12 Firing circuits. The piston actuator, and detonator firing circuits shall each provide a minimum of 1500 ergs to a 10 ohm load within a 20 millisecond interval when tested as specified in 4.5.5.12.

3.6.12.1 Piston actuator firing circuit. When functioned, the piston actuator firing circuit shall remain on for only one PTB clock count. The piston actuator firing circuit shall function again after the detonator circuit of 3.6.12.2 has turned off, and remain on as long as the electronics is functionally operable or is intentionally reset.

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3.6.12.2 Detonator firing circuit. When functioned, the detonator firing circuit shall remain on for 64 minimum and 128 maximum PTB clock counts. Thereafter, it shall remain off as long as the electronics is functionally operable or is intentionally reset.

3.7 Low temperature operation. The electronic assembly shall meet the performance characteristics specified in 3.6 at minus 40° - 5°F when tested as specified in 4.5.5.13.

3.8 High temperature operation. The electronic assembly shall meet the performance characteristics specified in 3.6 at 145° + 5°F when tested as specified in 4.5.5.14.

3.9 Soldering. Soldering procedures and acceptance criteria for the soldered connections shall comply with MIL-S-45743, MIL-S-46844, Appendix A, and as specified herein.

3.9.1 Solderability. Electronic assembly parts that are to be soldered shall comply with the solderability requirements of MIL-S-45743, MIL-S-46844, MIL-STD-202 (Method 208), and Appendix A herein.

3.10 Verification of safe condition of S&A assemblies prior to assembly to Test Mine Assemblies. Prior to assembly in the Test Mine Assembly, positive verification of the following safety features shall be made for each S A Assembly (See 4.5.7.1):

a. Presence and safe position of the interlock assembly.

b. Presence of the borerider ball lock.

c. Minimum dimension of borerider: distance from end of sleeve to borerider.

d. Safe position of spring loaded slider assembly: distance from end of S&A to slider assembly, max.

e. Safe position of slider release pin and shear pin.

3.11 Verification of safe condition of Electronic MCD Assemblies. Circuit resistance checks of each completed Electronic-MCD Assemblies shall be verified, as specified in 4.5.7.2.

3.12 Process control. The contractor shall submit to the government for approval with the first article sample, process

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control documents for the processes listed in the following subparagraphs. These processes shall be those used in the fabrication of the first article sample and shall not be changed after approval of the first article sample without prior approval of the government. At unscheduled intervals during the operation of the approved processes, the Government inspector shall observe the processes to ascertain that production to the approved process control documents is being maintained. No deviation without approval from the procuring activity shall be allowed. In the event that the contractor desires a change in an approved process control document, a new first article sample may, at the discretion of the Government be required.

3.12.1 Process control of molded plastic parts. A process control document shall be prepared for each of the molded plastic parts. These documents shall briefly describe the entire process and shall fully document the controls to be utilized for maintaining the process. They shall specify the time, temperature, pressure ranges and cure process which shall be maintained during production of the molded parts.

3.12.2 Process control of potting. A process control document shall be prepared for potting the electronic assembly. It shall fully describe the entire process and the controls for maintaining the process. It shall specify the preparation of all materials, pot life, and curing temperature and time.

3.12.3 Process control of soldering. Process control documents/procedures shall be prepared for all soldering operations. These documents shall fully describe the entire process and controls for maintaining the process/procedures in compliance with the soldering requirements of 3.9.

3.13 Material control. The contractor shall prepare and submit to the Government a material control document. This document shall detail the procedures used to identify and control the components/subassemblies to assure that only qualified components/subassemblies from qualified suppliers are used in the manufacture or assembly of the BLU-91 mine, Electronic-MCD Assembly.

3.14 Qualified suppliers. A qualified component supplier shall be one who has successfully met the requirements of this specification and First Article inspection specified herein.

3.15 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.16 Workmanship.

3.16.1 Mechanical workmanship. All parts and assemblies shall be fabricated and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

3.16.2 Electrical workmanship All soldered electrical connections shall comply with MIL-S-45743, MIL-S-46844, Appendix A herein, and requirements of applicable drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspction and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample, as designated by the Contracting Officer, for evaluation in accordance with the provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the items and sample quantities listed in Table I.

4.3.2 Inspections to be performed.

4.3.2.1 Components and subassemblies. see MIL-A-48078 and Table I specified herein.

4.3.2.2 Electronic assemblies (Dwg. 9330378). The electronic assemblies listed on Table I shall be serialized for identification and subjected to the following tests.

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4.3.2.2.1 Temperature cycle. Subject the electronic assemblies to temperature cycle per MIL-STD-202, Method 107D, Condition A1, except that the temperature range shall be minus 65°F to 160°F. (Non-destructive test).

4.3.2.2.2 Mechanical shock. Following temperature cycle, subject the electronic assemblies to mechanical shock per MIL-STD-202, Method 213B, Condition E with a 1000g peak and 2ms max. duration. (Non-destructive test).

4.3.2.2.3 Performance test. Following mechanical shock, the electronic assemblies performance characteristics shall be verified by testing as specified in 4.4.2.33. The test sample shall be rejected if, during test, a defect occurs. (Non-destructive test).

4.3.2.3 Test mine assemblies. The Electronic-MCD Assemblies complying with 4.4.2.34, manufactured from the units used in 4.3.2.2.3 above, and complying with 4.4.2.35 shall be assembled into test mines (Dwg. 9349483) and subjected to the functioning test of 4.3.2.4 (See 6.10).

4.3.2.4 Function tests. The test mine assemblies shall be subjected to transportation vibration in accordance with 4.5.3 and function tested in the mode and temperature noted in Figure 1. Each group shall be preconditioned at only one temperature for a minimum of four (4) hours without borerider clips installed. Within five (5) minutes after being removed from the conditioning chamber, the test mines shall have borerider clips installed and shall be launched to enable the S&A's to activate the battery cells and set the self-destruct feature for the desired SD time (4.4.6). (Destructive test).

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Sample Size	Function Mode	
	Sensor	Self-Destruct
84	48	36

Function		Function Temperature			Accept/Reject			
		-40°-5°F	Amb	145°+5°F	Failure or Dud	Early SD	Hazard-ous Dud	Premature Arming or Detonation
Sensor	-	16	16	16	0-1	0-1	0-1	0-1
Self-destruct	SD_1	4	4	4	0-1	0-1	0-1	0-1
	SD_2	4	4	4				
	SD_3	4	4	4				

NOTES:
1. See 6.9 for definitions of defects listed under accept-reject.
2. See 4.5.4.4 for data to be recorded.
3. See 4.5.4.1 for sensor function.
4. See 4.5.4.2 for S.D. function.

FIGURE 1. Function test mode.

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4.3.3 Rejection. See MIL-A-48078 and as specified herein.

4.3.4 Requalification. The contractor shall be subjected to another first article as stipulated in the contract if any of the following occur:

a. Manufacturing techniques or process controls which affect the form, fit, or function of the assembly are revised by either the contractor or one of his subcontractors supplying parts or assemblies or materials.

b. The contractor relocates his production facilities or portions thereof.

c. The contractor desires to use a new subcontractor (unqualified) for material, parts or assemblies.

d. A change occurs in the design, material, performance or assembly of the Electronic-MCD Assembly.

e. A lapse in production for a period of 90 days or as stipulated in the contract.

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-E-63511 (AR)

PARAGRAPH	TITLE: Electronic-MCD Assemblies		SHEET 1 OF 3		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Transistor, (JANTXV 2N2907) (Dwg. 9287122)				
	Examination for defects	10	--	3.2	4.4.2.4
	Transistor (JANTXV 2N2222A) (Dwg. 9287122)				
	Examination for defects	10	--	3.2	4.4.2.4
	Semiconductor Device, Diode (JANTX IN645)				
	Examination for defects	10	--	3.2	4.4.2.5
	Diode, Zener (JANTX IN4463)				
	Examination for defects	10	--	3.2	4.4.2.6
	Diode, Zener (JANTX IN4471)				
	Examination for defects	10	--	3.2	4.4.2.6
	Terminal (Dwg. 9292191)				
	Examination for defects	10	--	3.2	4.4.2.14
	Printed Wiring Board (Dwg. 9332389)				
	Examination for defects	20	--	3.2	4.4.2.16
	Solderability		--	3.2	4.4.2.16

NOTES:

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE: Electronic-MCD Assemblies		SHEET 2 OF 3		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Magnetometer Assembly (Dwg. 9287146) Examination for defects	10	--	3.2	4.4.2.20
	Shorting Bar, Ceramic (Dwg. 9292157) Examination for defects	10	--	3.2	4.4.2.21
	Shorting Bar and Connector Assembly (Dwg. 9292158) Examination for defects	10	--	3.2	4.4.2.22
	Receiver Coil Assembly (Dwg. 9292244) Examination for defects	10	--	3.2	4.4.2.24
	Band Printed Wiring (Dwg. 9292230) Examination for defects	10	--	3.2	4.4.2.26
	Coil (Dwg. 9292232) Examination for defects	10	--	3.2	4.4.2.27
	Lid, Electronic Assembly (Dwg. 9292243) Examination for defects	10	--	3.2	4.4.2.29
	Housing, Electronic Assembly (Dwg. 9332372) Examination for defects	10	--	3.2	4.4.2.30

NOTES:

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TABLE I. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-E-63511 (AR)

PARAGRAPH	TITLE: Electronic-MCD Assemblies		SHEET 3 OF 3		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Printed Wiring Board Assembly (Dwg. 9330377)				
	Examination for defects	10	--	3.2	4.4.2.32
	Electronic Assembly (Dwg. 9330378) 1/				
	Mechanical	84	--	3.2	4.4.2.34
	Performance Characteristics @ Ambient	84	--	3.2	4.4.2.33
	Performance Characteristics @ -40° -5°F	84	--	3.2	4.4.2.33
	Performance Characteristics @ +145° +5°F	84	--	3.2	4.4.2.33
	Electronic-MCD Assembly (Dwg. 9330379) 2/				
	Examination for defects	84	--	3.2	4.4.2.35

NOTES:

1/ Electronic Assemblies shall be used in Electronic-MCD Assemblies (Dwg. 9330379) in compliance with 4.3.2.3.

2/ Electronic-MCD Assemblies shall be function tested in accordance with 4.3.2.3 and 4.3.2.4.

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of electronic assemblies and sub-assemblies shall contain:

a. Batteries and/or power supply assemblies bearing the same lot interfix number from one manufacturer.

b. Triple operational amplifiers bearing the same lot interfix number from one manufacturer.

c. Integrated circuits, CMOS, sensors bearing the same lot interfix number from one manufacturer.

d. Integrated circuits, CMOS, timer bearing the same lot interfix number from one manufacturer.

e. Networks, capacitator-resistor, or resistor bearing the same lot interfix number from one manufacturer.

f. Electrical battery primers bearing the same lot interfix number from one manufacturer.

g. Fuses bearing the same lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Forms, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Normal Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Resistor, Fixed Composition, ±5%, 1/8 W	1 OF 1	MIL-R-39008/4 PN See Below
			NEXT HIGHER ASSEMBLY 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Part Number / Part Value RCR05G 30 5JS — 3 M ohm RCR05G 47 2JS — 4.7 K ohm RCR05G 10 2JS — 1 K ohm RCR05G 13 1JS — 130 ohm				
CRITICAL	None defined				
MAJOR					
101	Resistance		0.40%	3.2	MIL-R-39008/4
102	Solderability, 2 leads	10	10-0-1	3.9.1	MIL-STD-202, Method 208
MINOR					
201	Identification or marking incorrect		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287099 P/N 9287099-1611
4.4.2.2	Resistor, Fixed Film				NEXT HIGHER ASSEMBLY 9330377
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Resistance	10	0.40%	3.2	
102	Solderability	10	10-0-1	3.9.1	MIL-STD-202, Method 208
MINOR					
201	Identification and marking		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE:		SHEET 1 OF 1		DRAWING NUMBER 9287099 P/N 9287099-1/
4.4.2.3	Resistor, Fixed Film				NEXT HIGHER ASSEMBLY 9330377
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Solderability	10	10-0-1	3.9.1	MIL-STD-202, Method 208
MINOR					
201	Identification and marking		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

1/ Select at assembly.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Transistor, (P/N 9287122-1, JANTXV 2N2907 and 9287122-2 JANTXV 2N 2222A)	1 OF 1	9287122
			NEXT HIGHER ASSEMBLY 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Wire material	32	1/	3.2	Visual
102	Wire bond	32	1/	3.2	Pull tester 2/
103	Compliance with MIL-S-19500/291 or/255	--	--	3.2	3/ Certification
104	Solderability (3 leads ea.)	10	10-0-1	3.9.1	MIL-S-45743
105	Leakage current Icer over max.		0.40%	3.2	4/
106	Internal visual	32	1/	3.2	MIL-STD 750, Method 2072
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

1/ Reject for one or more leads defective.
2/ Pull all wires on each device.
3/ Visual evidence of JANTXV marking.
4/ See applicable note and test schematic of dwg.

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Semi-conductor Device, Diode, Type JANTX IN645-1	1 OF 1	9287168-2
			NEXT HIGHER ASSEMBLY: 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Missing or incorrect color band on cathode end of diode		100%	3.2	Electrical, 1/
2	Reverse current		100%	3.2	MIL-STD-750, Method 4016, dc
MAJOR					
101	Forward voltage		100%	3.2	MIL-STD-750, Method 4011
102	Solderability (2 leads)	10	10-0-1	3.9.1	MIL-S-45743
103	Conformance to MIL-S-19500/240			3.2	2/ Certification
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES: 1/ Color band end shall be used as cathode end during reverse current test; reverse current test is used to confirm that the color band is on cathode end of diode.
2/ Visual evidence of JANTX marking.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Diode, Zener, JANTX IN 4463 and JANTX IN4471	1 OF 1	MIL-S-19500/406
			NEXT HIGHER ASSEMBLY 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Missing or incorrect color band on cathode end of diode		100%	3.2	Electrical, 1/
2	Reverse current		100%	3.2	MIL-STD-750, Method 4016, dc
3	Zener breakdown voltage Vz		100%	3.2	MIL-STD-750, Method 4022
MAJOR					
101	Forward voltage drop		100%	3.2	MIL-STD-750, Method 4011
102	Solderability (2 leads)	10	10-0-1	3.9.1	MIL-S-45743
103	Conformance to MIL-S-19500/406			2/	2/ Certification
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:
1/ Color band end shall be used as cathode end during reverse current test; reverse current test is used to confirm that the color band is on cathode end of diode.
2/ Visual evidence of JANTX marking.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Diode, Hot Carrier (JAN IN5711 MIL-S-19500/127)				9287169
					NEXT HIGHER ASSEMBLY 9330377
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Missing or incorrect color band on cathode end of diode		0.40%	3.2	Electrical, 1/
102	Reverse current over max.		0.40%	3.2	MIL-STD-750, Method 4016, dc
103	Forward voltage drop		0.40%	3.2	MIL-STD-750, Method 4011
104	Solderability	10	10-0-1	3.9.1	MIL-S-45743
105	Evidence of JAN marking		--	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES: 1/ Color band end shall be used as cathode end during reverse current test; reverse current test is used to confirm that the color band is on cathode end of diode.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Capacitor, Fixed, Plastic Dielectric (.22 uF)	1 OF 1	9287104
			NEXT HIGHER ASSEMBLY 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 30
102	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 30
103	Insulation resistance		0.40%	3.2	MIL-STD-202, Method 30
104	Solderability (2 leads)	10	10-0-1	3.9.1	MIL-S-45743
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.9	Capacitor, Fixed, Electrolytic Tantalum, Solid (82 uF)	1	1	9287126-2
				NEXT HIGHER ASSEMBLY: 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Breakdown voltage	10	10-1-2	3.2	Dwg. 9292220
105	Internal workmanship	20	20-1-2	3.2	Dwg. 9292220
106	Weld strength (2 sides)	8	8-0-1	3.2	Pull Tester
107	Solderability (2 leads)	10	10-0-1	3.9.1	MIL-S-45743
108	Sleeve missing		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Capacitor, Fixed, Electrolytic Tantalum, Solid	1 OF 1	9287096 1/
			NEXT HIGHER ASSEMBLY: 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Breakdown voltage	10	10-1-2	3.2	Dwg. 9292220
105	Internal workmanship	20	20-1-2	3.2	Dwg. 9292220
106	Weld strength (2 sides)	8	8-0-1	3.2	Pull Tester
107	Solderability (2 leads)	10	10-0-1	3.9.1	MIL-S-45743
108	Sleeve missing		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

1/. Applies to P/N 9287096-1, -2, -4, -6, -10, -11.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Capacitor, Ceramic, Fixed Dielectric Molded, Axial				MIL-C-39014/5
					NEXT HIGHER ASSEMBLY 9330377
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Solderability, (2 leads)	10	10-0-1	3.9.1	MIL-S-45743
105	Conformance to MIL-C-39014/5		--	3.2	Certification
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Capacitor, Ceramic, Fixed Dielectric-Axial Lead	1 OF 1	9292151 P/N 9292151-1
			NEXT HIGHER ASSEMBLY: 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 30
102	Leakage current		0.40%	3.2	MIL-STD-202, Method 30
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 30
104	Solderability, (2 leads)	10	10-0-1	3.9.1	MIL-S-45743
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Eyelet				9292146
					NEXT HIGHER ASSEMBLY 9330377
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Solderability	20	20-0-1	3.9.1	MIL-STD-202, Method 208
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Terminal, Stud				9292191
					NEXT HIGHER ASSEMBLY 9330377
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Solderability	20	20-0-1	3.9.1	MIL-STD-202, Method 20
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.15	Support Foam		9292197
			NEXT HIGHER ASSEMBLY
			9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Adhesive missing		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Board, Printed Wiring	1 OF 1	9332389
			NEXT HIGHER ASSEMBLY: 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	O.D. on board oversize		0.40%	3.2	Gage
102	True position of O.D. to datum "A"		0.40%	3.2	Gage
103	Center hole diameter undersize		0.40%	3.2	Gage
104	True position of C. diam. with datum "A"		0.40%	3.2	Gage
105	Large circular cutout, 2 places		0.40%	3.2	Gage
106	Diameter of component terminal holes		0.40%	3.2	Gage
107	Circuits, discontinuous		100%	3.2	Tester
108	Solderability	10	10-0-1	3.9.1	MIL-S-46844
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.17	Fuse, Electric		9328688
			NEXT HIGHER ASSEMBLY 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Fuse resistance		100%	3.2	Milliohmmeter, 2/
102	Case to lead insulation resistance		100%	3.2	Megohmmeter
103	No fire current for 10 second period, min.	1/	0.40%	3.2	Tester
104	Bridge wire material		100%	3.2	Certification
105	Fuse dimensioning		0.40%	3.2	Gages
106	All fire energy (capacitor discharge) test		0.40%	3.2	Tester
107	Solderability	10	10-0-1	3.9.1	MIL-S-45743
108	Evidence of etching compound or contaminants	3/	--	3.2	Chemical analysis
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

1/ Use these units for Major 106: All fire energy test.
2/ Limit test current to 1 ma.
3/ One (1) foot from each spool of bridge wire material.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Microcircuit, Digital, CMOS	1 OF 1	9287124
			NEXT HIGHER ASSEMBLY 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Wire bond strength	5	1/	3.2	Pull tester
102	Wire material incorrect	5	5-0-1	3.2	Visual
103	Solderability (all leads)	5	5-0-1	3.9.1	MIL-S-45743
104	Electrical characteristics		--	3.2	Certification
105	Screening		--	3.2	Certification
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

1/ Pull wires on each device - reject for one (1) or more bond failures.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Core, Toroid				9287094
					NEXT HIGHER ASSEMBLY 9292216
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Internal construction (see applicable note of dwg.)	32	32-0-1	3.2	Visual and microscope
102	Weld strength	32	32-0-1	3.2	Visual and manual
103	Constant-current Flux reset test		100%	3.2	Test equipment
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.20	Magnetometer Assembly	1 OF 1	9287146
			NEXT HIGHER ASSEMBLY: 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Evidence of coil unwinding, or self-bonding missing		0.40%	3.2	Visual
102	Improper conformal coating of winding		0.40%	3.2	Visual
103	Leads improperly soldered to terminals		0.40%	3.2	Visual
104	Resistance improper		0.40%	3.2	MIL-STD-202, Method 303
105	Sensitivity @ Z=0 Oersted; Ambient		0.40%	3.2	Dwg. 9287149
106	Linearity @ Z=+0.45 and -0.45 Oersted; Ambient		0.40%	3.2	Dwg. 9287149
107	Sensitivity @ Z=0; Temperature extremes		0.40%	3.2	Dwg. 9287149
108	Impedance improper		0.40%	3.2	Dwg. 9287149
109	Solderability (2 leads)	20	20-0-1	3.9.1	MIL-STD-202, Method 208
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.21	Shorting Bar, Ceramic				9292157
					NEXT HIGHER ASSEMBLY
					9330377
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Thickness		0.40%	3.2	Gage
102	Solderability	10	10-0-1	3.9.1	MIL-STD-202, Method 208
MINOR					
201	Total height		0.65%	3.2	Gage
202	Total length		0.65%	3.2	Gage
203	Dielectric glaze missing or incomplete		0.65%	3.2	Visual
204	Solder coating outside area "A" incorrect		0.65%	3.2	Visual
205	Thickness of solder pad, 2 places		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292158
4.4.2.22	Shorting Bar and Connector Assembly				NEXT HIGHER ASSEMBLY 9330377
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Solder workmanship incorrect		0.40%	3.16	Visual (10X)
102	Evidence of cracks or chips		0.40%	3.2	Visual
MINOR					
201	Edge of connector extends beyond ceramic, 2 places		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Standoff, EBP	1 of 1	9292171
			NEXT HIGHER ASSEMBLY 9292231

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Diameter for EBP lead, 2 places		0.40%	3.2	Gage
102	True position of C.D. 101 Diam. with Datum A		0.40%	3.2	Gage
MINOR					
201	Width of groove		0.65%	3.2	Gage
202	Outside diameter		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.24	Receiver Coil Assembly				9292244
					NEXT HIGHER ASSEMBLY 9292231
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Polarity (direction of turns)		100%	3.2	Electrical
MAJOR					
101	Spacing of coil's 1st turn to edge of circuit board, min. - 3 places		0.40%	3.2	Gage
102	Solder workmanship, 2 places		0.40%	3.16	Visual
103	Thickness, max.		0.40%	3.2	Gage
104	Spacing of coil to solder pad, min. - 4 places		0.40%	3.2	Gage
105	Resistance		0.40%	3.2	Milliommeter
106	Inductance		0.40%	3.2	Impedance Bridge
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.25	MCD Assembly				9292231
					NEXT HIGHER ASSEMBLY 9330379
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Evidence of poor soldering		0.40%	3.16	Visual
MINOR					
201	Protrusion of lead wires, max.		0.65%	3.2	Gage
202	Wrong end of standoff adjacent to primer		0.65%	3.2	Visual
203	Height of primer above coil assembly		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292230
4.4.2.26	Board, Printed Wiring				NEXT HIGHER ASSEMBLY 9292244
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Largest outside diameter		0.40%	3.2	Gage
102	Total thickness, max.		0.40%	3.2	Gage
103	Diameter of "A" hole, 4 places		0.40%	3.2	Gage
104	Location of "A" hole, 4 places		0.40%	3.2	Gage
105	Diameter of "B" hole, 2 places		0.40%	3.2	Gage
106	Location of "B" hole, 2 places		0.40%	3.2	Gage
107	Profile incorrect		0.40%	3.2	Gage
108	Conductor width and spacing incorrect		0.40%	3.2	Gage
MINOR					
201	Plating of pad incorrect, 7 places		0.65%	3.2	MIL-STD-275
202	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292232
4.4.2.27	Coil				NEXT HIGHER ASSEMBLY 9292244
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Outside diameter, max.		0.40%	3.2	Gage
102	Inside first coil diameter, min.		0.40%	3.2	Gage
103	Thickness, max.		0.40%	3.2	Gage
MINOR					
201	Length of flat, 2 places		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.28	Bushing, Power Supply	1 OF 1	9292172
			NEXT HIGHER ASSEMBLY: 9330378

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Inside diameter		0.65%	3.2	Gage
202	True position of I.D.		0.65%	3.2	Gage
203	Outside diameter		0.65%	3.2	Gage
204	Chamfer missing, one end		0.65%	3.2	Visual
205	Evidence of poor workmanship.		0.65%	3.16	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.29	Lid, Electronics Assembly		9292243
			NEXT HIGHER ASSEMBLY 9330378

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined	1/			
MAJOR					
101	Profile of MCD assembly nest		0.40%	3.2	Gage
102	Location of MCD assembly nest with datum "A"		0.40%	3.2	Gage
103	Height of lid		0.40%	3.2	Gage
104	Outside diameter		0.40%	3.2	Gage
MINOR					
201	Inside diameter		0.65%	3.2	Gage
202	Diameter of terminal hole, 7 places		0.65%	3.2	Gage
203	Height of boss (for MCD) 2 places		0.65%	3.2	Gage
204	True position of outside diameter to datum "A"		0.65%	3.2	Gage
205	Diameter of EBP clearance hole		0.65%	3.2	Gage
206	True position of EBP clearance hole		0.65%	3.2	Gage
207	Thickness at base of MCD assembly nest		0.65%	3.2	Gage
208	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

1/ See 4.5.1 for inspection of mold features.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.30	Housing, Electronics Assembly				9332372
					NEXT HIGHER ASSEMBLY 9330378
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined	1/			
MAJOR					
101	Height to "D" hole flat (-B-)		0.40%	3.2	Gage
102	Radius of MDF cord groove, 7 places		0.40%	3.2	Gage
103	True position of outside diameter to datum "A"		0.40%	3.2	Gage
104	Outside diameter		0.40%	3.2	Gage
105	Overall width		0.40%	3.2	Gage
MINOR					
201	Height to clearance flat (for cable)		0.65%	3.2	Gage
202	Spherical radius		0.65%	3.2	Gage
203	"D" hole diameter		0.65%	3.2	Gage
204	Width of test point hole		0.65%	3.2	Gage
205	Height of test point cavity		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

1/ See 4.5.1 for inspection of mold features.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9330379
4.4.2.31	Electronics - MCD Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Conductive foam missing from terminals		0.40%	5.1	Visual
102	Barrier bag missing or damaged		0.40%	5.1	Visual
103	Marking improper		0.40%	5.3	Visual
MINOR					
201	Evidence of poor workmanship		0.65%		Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.32	Printed Wiring Board Assembly				9330377
					NEXT HIGHER ASSEMBLY
					9330378
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Terminals: loose, bent, or canted more than 5 degrees		0.40%	3.2	Visual/Gage
102	Any component damaged, or missing, or improperly assembled		0.40%	3.2	Visual
103	Component(s) height from board (max)		0.40%	3.2	Gage
104	Length of leads/solder on bottom of board		0.40%	3.2	Gage
105	Soldering workmanship		0.40%	3.16	Visual
106	Conformal coat missing from noted components		0.40%	3.2	Visual
107	Shorting bar support foam missing		0.40%	3.2	Visual
108	Shorting bar assembly not properly seated in cavity		0.40%	3.2	Visual
109	Air-gap present between support foam and cavity		0.40%	3.2	Visual
110	Conformal coat in noted non-conformal coat areas		0.40%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 6		DRAWING NUMBER 9330378
4.4.2.33	Electronic Assembly (Performance Characteristics) (Ambient)				NEXT HIGHER ASSEMBLY 9330379
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Primary Time Base (PTB) oscillator period, over max. or under min.		100%	3.6.2	4.5.5.2
2	Test Time Base (TTB) oscillator period, over max.		100%	3.6.3	4.5.5.3
3	Oscillator check circuit does not detect a PTB period that is greater than the TTB period		100%	3.6.11	4.5.5.11
4	PTB clock count at SD1 self-destruct signal, over max.		100%	3.6.7	4.5.5.9
5	PTB clock count AT SD2 self-destruct signal, over max.		100%	3.6.8	4.5.5.8
6	PTB clock count SD3 self-destruct signal, over max.		100%	3.6.9	4.5.5.7
7	PTB clock count at S&A arm signal, under min.		100%	3.6.4	4.5.5.4
8	One or more fuse open		100%	3.2	Test
9	Blocking or zener diodes have improper polarity; breakover voltage, open/short circuit (SD set circuit)		100%	3.2	Test

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.33	Electronic Assembly (Performance Characteristics) (Ambient)	2 OF 6	9330378
			NEXT HIGHER ASSEMBLY 9330379

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MAJOR					
101	Quiescent current		100%	3.6.1	4.5.5.1
102	PTB clock count at electronic arm incorrect		100%	3.6.5	4.5.5.5
103	Low voltage detect, or magnetic sensor signal not inhibited before sensor enable		100%	3.6.5	4.5.5.5
104	Low voltage detect circuit trigger level		100%	3.6.6	4.5.5.6
105	Magnetic sensor function test		100%	3.6.10	4.5.5.10
106	Piston actuator firing circuit energy output		100%	3.6.12	4.5.5.12
107	Detonator firing circuit energy output		100%	3.6.12	4.5.5.12
108	PTB clock count at SD1 self-destruct signal, under min.		100%	3.6.7	4.5.5.9
109	PTB clock count at SD2 self-destruct signal, under min.		100%	3.6.8	4.5.5.8
110	PTB clock count at SD3 self-destruct signal, under min.		100%	3.6.9	4.5.5.7
111	Arming signal not inhibited		100%	3.6.4.1	4.5.5.4.1
MINOR	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 3 OF 6		DRAWING NUMBER
4.4.2.33	Electronic Assembly (Performance Characteristics) (minus 40°-5°F)				9330378
					NEXT HIGHER ASSEMBLY 9330379
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Primary Time Base (PTB) oscillator period, over max. or under min.	1/	0-1	3.6.2	4.5.5.2
2	Test Time Base (TTB) oscillator period, over max.		0-1	3.6.3	4.5.5.3
3	Oscillator check circuit does not detect a PTB period that is greater than the TTB period		0-1	3.6.11	4.5.5.11
4	PTB clock count at SD1 self-destruct signal, over max.		0-1	3.6.7	4.5.5.9
5	PTB clock count at SD2 self-destruct signal, over max.		0-1	3.6.8	4.5.5.8
6	PTB clock count SD3 self-destruct signal, over max.		0-1	3.6.9	4.5.5.7
7	PTB clock count at S&A arm signal, under min		0-1	3.6.4	4.5.5.4
8	One or more fuse open		0-1	3.2	Test
9	Blocking or zener diodes have improper polarity; breakover voltage, open/ short circuit (SD set circuit)		0-1	3.2	Test

NOTES:

1/ Sample size for critical characteristics to be the same as determined for major characteristics. No defects allowed for critical characteristics.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 4 OF 6	DRAWING NUMBER
4.4.2.33	Electronic Assembly (Performance Characteristics) (minus 40°-5°F)		9330378
			NEXT HIGHER ASSEMBLY 9330379

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Quiescent current		0.40%	3.6.1	4.5.5.1
102	PTB clock count at sensor enable incorrect		0.40%	3.6.5	4.5.5.5
103	Low voltage detect, or magnetic sensor signal not inhibited before sensor enable		0.40%	3.6.5	4.5.5.5
104	Low voltage detect circuit trigger level		0.40%	3.6.6	4.5.5.6
105	Magnetic sensor function test		0.40%	3.6.10	4.5.5.10
106	Piston actuator firing circuit energy output		0.40%	3.6.12	4.5.5.12
107	Detonator firing circuit energy output		0.40%	3.6.12	4.5.5.12
108	PTB clock count at SD1 self-destruct signal under min.		0.40%	3.6.7	4.5.5.9
109	PTB clock count at SD2 self-destruct signal under min.		0.40%	3.6.8	4.5.5.8
110	PTB clock count at SD3 self-destruct signal under min.		0.40%	3.6.9	4.5.5.7
111	Arming signal not inhibited		0.40%	3.6.4.1	4.5.5.4.1
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 5 OF 6		DRAWING NUMBER
4.4.2.33	Electronic Assembly (Performance Characteristics) (145°+5°F)				9330378
					NEXT HIGHER ASSEMBLY 9330379
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Primary Time Base (PTB) oscillator period, over max. or under min.	1/	0-1	3.6.2	4.5.5.2
2	Test Time Base (TTB) oscillator period, over max.		0-1	3.6.3	4.5.5.3
3	Oscillator check circuit does not detect a PTB period that is greater than the TTB period		0-1	3.6.11	4.5.5.11
4	PTB clock count at SD1 self-destruct signal, over max.		0-1	3.6.7	4.5.5.9
5	PTB clock count at SD2 self-destruct signal, over max.		0-1	3.6.8	4.5.5.8
6	PTB clock count SD3 self-destruct signal over max.		0-1	3.6.9	4.5.5.7
7	PTB clock count at S&A arm signal, under min.		0-1	3.6.4	4.5.5.4
8	One or more fuse open		0-1	3.2	Test
9	Blocking or zener diodes have improper polarity; breakover voltage, open/short circuit (SD set circuit)		0-1	3.2	Test

NOTES: 1/ Sample size for critical characteristics to be the same as determined for major characteristics. No defects allowed for critical characteristics.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 6 OF 6		DRAWING NUMBER
4.4.2.33	Electronic Assembly (Performance Characteristics) (145°+5°F)				9330378
					NEXT HIGHER ASSEMBLY 9330379
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Quiescent current		0.40%	3.6.1	4.5.5.1
102	PTB clock count at sensor enable incorrect		0.40%	3.6.5	4.5.5.5
103	Low voltage detect, or magnetic sensor signal not inhibited before sensor enable		0.40%	3.6.5	4.5.5.5
104	Low voltage detect circuit trigger level		0.40%	3.6.6	4.5.5.6
105	Magnetic sensor function test		0.40%	3.6.10	4.5.5.10
106	Piston actuator firing circuit energy output		0.40%	3.6.12	4.5.5.12
107	Detonator firing circuit energy output		0.40%	3.6.12	4.5.5.12
108	PTB clock count at SD1 self-destruct signal under min.		0.40%	3.6.7	4.5.5.9
109	PTB clock count at SD2 self-destruct signal under min.		100%	3.6.8	4.5.5.8
110	PTB clock count at SD3 self-destruct signal under min		0.40%	3.6.9	4.5.5.7
111	Arming signal not inhibited		0.40%	3.6.4.1	4.5.5.4.1
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9330378
4.4.2.34	Electronics Assembly (Mechanical)				NEXT HIGHER ASSEMBLY 9330379
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Excessive voids/bubbles in potting		0.40%	3.3.2	4.5.2
102	Potting material characteristics		1/	3.3.2	4.4.3.1
103	Power supply bushing above surface "A"		0.40%	3.2	Visual
104	Terminal pins and pads are not clean		0.40%	3.2	Visual
105	Overall height, max.		0.40%	3.2	Gage
106	Improper mating of electronic lid and housing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

1/ Sample in accordance with 4.4.3.1. Potting process shall be verified daily.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.35	Electronics-MCD Assy. (Prior to Assy. of Test Mines	1 OF 1	9330379
			NEXT HIGHER ASSEMBLY
			- - - - -

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Resistance and voltage checks not in compliance with Table II 1/		100%	3.11	4.5.7.2
Major					
101	Terminal pins not clean		0.40%	3.2	Visual
102	Visual damage		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

1/ Data item description DI-R-1721 applies. Submission of the report shall be whenever defects are found.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.36	Electronics-MCD Assembly		9330379
			NEXT HIGHER ASSEMBLY
			- - - - -

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	Lot Acceptance Tests	1/			
1	Unsafe after Transportation-Vibration		0-1	3.4.1	4.5.3
2	Impact failure		0-1	3.5	4.5.4
3	Premature arming, detonation		0-1	3.5	4.5.4
4	Hazardous dud		0-1	3.5	4.5.4
Major	Performance Test: Launch/Function	1/			
101	Sensor or SD Function		--	3.5	4.5.4.1
102	Early SD		--	3.5	4.5.4

NOTES:

1/ See sheet 2.

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QUALITY CONFORMANCE INSPECTION

Paragraph 4.4.2.36

Dwg. 9330379

Inspection Lot Size	Sample Size	Prime Sensor	SD
501-1200	36	9	27
1201-3200	60	15	45
3201-10000	84	21	63

TEMPERATURE CONDITIONING													ACCEPT-REJECT CRITERIA 2/			
	-40-5°F				Ambient				145+5°F							
Sample Size	Sensor	SD 1	SD 2	SD 3	Sensor	SD 1	SD 2	SD 3	Sensor	SD 1	SD 2	SD 3	Failure 1/ Dud	Early SD	Hazardous Dud	Premature Det
36	3	3	3	3	3	3	3	3	3	3	3	3	1-2	2-3	0-1	0-1
60	5	5	5	5	5	5	5	5	5	5	5	5	2-3	4-5	0-1	0-1
84	7	7	7	7	7	7	7	7	7	7	7	7	3-4	6-7	0-1	0-1

1/ Includes sensor and SD failures.
2/ See 6.9 for definitions of defects.
3/ See 4.5.4.4 for data to be recorded.
4/ See 4.5.4.1 for sensor function.
5/ See 4.5.4.2 for SD function.

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4.4.3 Testing. Testing is described in the First Article, Quality Conformance Inspection "Classification of Defects and Tests" forms, in 4.5, and as follows.

4.4.3.1 Potting material testing. The potting material shall be checked for tensile strength, and hardness in accordance with Dwg. 9287148. Two test samples shall be molded per the drawing for each batch of mixed material or each 1,000 electronic assemblies assembled during continuous production, or each day's production, whichever comes first. One test sample will be used for testing and one retained per the drawing. Identification shall be maintained between the sample and assemblies. Failure of the test specimen to comply with the requirements shall result in rejection of the lot of electronic assemblies assembled from that batch of potting material.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in Classification of Defects and Tests forms beginning with paragraph 4.4.2.1. The Contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and section 6.3 herein.

4.4.5 Test mine assembly. A test mine (Drawing 9349483) having the equivalent testing features of a regular mine but containing inert filler, booster, and clearing charges will be used for lot acceptance and first article sample test of the Electronics-MCD Assembly (see 6.10).

4.4.6 Launcher simulator. An air gun shall be used as a launcher for lot acceptance testing. Design to be submitted in accordance with 6.3. The air gun shall have the following features:

a. Impart sufficient velocity to the mines to provide impacts of 150 + 6.0 feet per second (fps).

b. Provide for borerider release after ejection from the muzzle but prior to impact, (launch with borerider clip installed).

c. Provide the interface with the SD setter to initiate the mine prior to/at launch.

4.4.6.1 SD setter. The SD setter shall provide the MCD signal necessary to initiate and set the proper SD time into the

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mine. The signals necessary to perform this function are contained on Drawing 1183AS616, Standard Waveform. Design to be submitted in accordance with 6.3.

4.4.7 Impact medium. The impact or target material will simulate an asphalt surface and will consist of three (3) 1.5 inch thick particle boards (Chipboard), with a density of 30 ± 2 pounds per cubic foot, with a 1/2 in. thick steel back-up plate.

4.5 Test methods and procedures.

4.5.1 Dimensional control of molded or plastic parts. In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (See Note 1), a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally (all dimensions on drawing) to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.4.2.1 through 4.4.2.36, after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. If any defective parts are found during qualification of the cavity, the cavity producing the defective part(s) will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part(s) shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request. Contractor designs of gages and test equipment required to perform the inspections listed herein shall be submitted in accordance with paragraph 4.4.4 and section 6.3 herein for approval prior to manufacture of equipment. The noted sub-paragraphs identify those items and minimum inspections subject to the requirements of this paragraph.

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NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. minutes) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be submitted in accordance with 6.5. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

NOTE 2: Those features of a part which are subsequently machined or otherwise worked upon to meet its drawing requirements shall be inspected in accordance with the classification of defects.

4.5.2 Potted electronic assembly. The potted electronic assembly shall not show evidence of voids, bubbles and delaminations exceeding the following criteria:

a. Bubbles within the sprue vent area are acceptable provided they do not contact the top or bottom surface of the printed wiring board (PWB). A bubble adjacent to a diode or metal film resistor may not contact the component for a length greater than one-quarter (1/4) of the diameter of that component. Bubbles in the remaining assembly are acceptable, provided they are less than one-quarter (1/4) inch in their maximum physical dimension and do not contact more than 5% of the surface area of any electrical component.

b Bubbles exposing any portion of a lead termination or junction are not acceptable.

c. Delaminations of the potting and the PWB are not acceptable.

d. Sprue cutoff shall be flush to .04 inches below the surface.

4.5.3 Transportation-Vibration. The test samples (without borerider clip installed) shall be serialized and secured in fixtures to a vibrator and tested in accordance with Test 119, Procedure 2, of MIL-STD-331. Observation shall be made for compliance with the requirement of 3.4.1 (see 6.9). Samples shall be used in subsequent testing.

4.5.4 Functional test of Electronic-MCD Assemblies. After completion of the transportation-vibration the test sample (without borerider clip installed) shall be divided into the

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required groups for temperature conditioning. Each group shall be subjected to only one temperature for a minimum period of four (4) hours as specified in 3.4.2. The test assemblies shall be taken from the conditioning chamber and tested as follows: Within five (5) minutes after being removed from the conditioning chamber, the test samples shall be launched to enable the S&A's to activate the battery cells and set the self-destruct feature for the desired SD time (see 4.4.6). After enable time has occurred, the test samples shall be functioned by sensor or SD modes. Enable time is defined as the maximum Electronic Arm Time specified on Dwg. 9332388, plus one (1) hour, minimum.

4.5.4.1 Sensor mode functioning. Following launch (see 4.5.4), and after enable time has occurred, the samples selected for sensor tests shall be functioned as follows:

a. Subject the test items to the Test Mine Assembly Test signature of Dwg. 9332388.

b. If no detonation is detected, and after a minimum of five (5) minutes have elapsed, attempt to function the test item a second time by repeating a. above.

c. If no detonation is detected on the second attempt the test sample shall be classed defective for sensor function.

d. Failing c. monitor the test sample until expiration of the selected SD time (see 4.5.4.2).

e. Observation shall be made for compliance with 3.5 and Dwg. 9332388.

f. Subsequent to the test and observing necessary safety precautions, the tested items shall be dismantled and failure analyzed (See 6.9). For duds, allow a min. of 24 hours after SD3 time out (see 4.5.4.2). (Destructive test).

4.5.4.2 Self-Destruct (SD) Mode Functioning. The test samples selected for SD mode (set at launch), shall be functioned as follows:

a. Monitor the test sample until expiration of the selected SD time.

b. Observation shall be made for compliance with 3.5 and Dwg. 9332388.

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c. If no detonation is detected by the expiration of the selected SD time the test sample shall be classed defective for SD function (Dud).

d. Subsequent to the test and observing necessary safety precautions, the tested items shall be dismantled and failure analyzed (see 6.9). Allow a minimum of 24 hours after SD_3 time out. (Destructive test)

WARNING: Test samples used in SD Functioning Test have active sensors, movement or magnetic disturbance will cause function by sensor mode.

4.5.4.3 Disposition of duds. Procedural plans or methods for disposing of duds and expended test items by either safely dismantling and failure analysis, or by safely destroying shall be furnished ARRADCOM for approval see 6.6.

Caution: When depleted, battery cells become unstable and generate high internal pressures.

4.5.4.4 Recording of functional data. The following functional data shall be recorded with serial numbers:

a. Orientation of test mine assemblies.

b. Time to arm (MCD input to actuation of micropiston actuator).

c. The number of main charge leads that functioned.

d. The number of attempts required for sensor function.

e. Time from MCD input (launch) to SD function.

f. The number of duds.

g. The number of hazardous duds (see 6.4).

h. After failure analysis, record cause of malfunction.

4.5.5 Electronic assembly performance tests. Increasing the PTB oscillator frequency to reduce test time is acceptable provided the higher frequency oscillator signal maintains an approximate 50% duty cycle and has a peak amplitude frequency oscillator signal which does not cause or mask malfunctions in the functional operation of the electronic assembly; the higher frequency oscillator signal is well within the capability of the test equipment to monitor and control it with an accuracy of one cycle; the actual procedure used to increase the frequency of the

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signal is approved by the government before it is implemented. When the PTB oscillator frequency is increased, the TTB oscillator frequency is to be proportionately increased or the TTB oscillator inhibited, as applicable, for the test being implemented. All testing shall be performed, unless otherwise specified at a supply voltage of 6.50 ± 0.15 volts, an ambient temperature of 70° + 5°F, and a relative humidity of 60 + 20%. Also, the rise time of the supply voltage applied to the electronic assembly under test shall not be less than 5 milliseconds nor greater than 50 milliseconds. The following tests (4.5.5.7 through 4.5.5.9) are to be performed in sequence.

4.5.5.1 Quiescent current test. With the electronic assembly under test connected to the test fixture and with the piston actuator, and the detonator terminals each connected to the ground through a 10 ohm resistor, measure the quiescent current of the electronic assembly. The measured current shall be observed for compliance with the requirement specified in 3.6.1. (Non-destructive test).

4.5.5.2 Primary Time Base (PTB) oscillator period test. With the electronic assembly under test connected to the test fixture and with both the primary and test time base circuits oscillating at their natural frequency, measure the period of the PTB oscillator at test point. Observe the measured period for compliance with the requirements specifed on Drawing 9332388 (See 3.6.2). (Non-destructive test).

4.5.5.3 Test Time Base (TTB) oscillator period test. With the electronic assembly under test connected to the test fixture and with both the primary and test time base oscillators operating at their natural frequency, measure the period of the TTB oscillator at test point. Observe the measured period for compliance with the requirements specified in Drawing 9332388. (See 3.6.3). (Non-destructive test).

4.5.5.4 S&A assembly arm signal test. With the electronic assembly under test connected to the test fixture, verify that the S&A assembly arm signal occurs at the PTB clock count specified on Dwg. 9332388 (See 3.6.4). This test requires that both oscillators be operating at their natural frequency. (Non-destructive test).

4.5.5.4.1 S&A assembly arm signal inhibit test. With the electronic assembly under test connected to the test fixture, and the TTB clock not running verify that the S&A arming signal does not occur at the PTB clock count specified on Dwg. 9332388. (see 3.6.4.1). (Non-destructive test).

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4.5.5.5 Electronic arm test. With the electronic assembly under test connected to the test fixture, verify that electronic arming occurs at the PTB clock count specified on Drawing 9332388 (See 3.6.5). Make the verification by incorporating, as a minimum, the following checks in the test procedure:

a. At the PTB clock count of 3 (±2) less than that required for electronic arm, activate magnetic sensor circuit and confirm that the detonator firing circuit does not turn on.

b. At the PTB clock count of 3 (±2) greater than that required for electronic arm, activate magnetic sensor circuit and confirm that the detonator firing circuit is turned on. (Non-destructive test).

4.5.5.6 Low voltage detect test. With the electronic assembly under test connected to the test fixture, verify that the low voltage detect circuit initiates the turn on of the detonator firing circuit when the supply voltage is reduced to the low voltage detect circuit trigger level specified on Dwg. 9332388 (see 3.6.6). For this test it is required that electrical arming has occurred. The initial supply voltage shall be 6.5 ± 0.15 volts and linearly reduced at a rate of 0.40 (max) volts/second until the low voltage detect circuit is triggered. Observation shall be made for compliance with the applicable requirement. (Non-destructive test.)

4.5.5.7 SD3 self-destruct signal test. With the electronic assembly under test connected to the test fixture and the fuses simulated open by connecting TP11 to the positive voltage supply, verify that the SD3 self-destruct signals, that initiate the turn on of the detonator firing circuit, occur at Pin A and at the PTB clock count specified on Dwg. 9332388 (see 3.6.9). After 64 min., 128 max., additional PTB clock counts verify that the detonator firing signal (Pin A) turns off and the PA firing signal (Pin B) turns on. This test is to be conducted before the SD2 self-destruct test (4.5.5.8). (Non-destructive test).

4.5.5.8 SD2 self-destruct signal test. With the electronic assembly under test connected to the test fixture, and the fuses simulated open by connecting TP12 to the positive voltage supply, verify that the SD2 self-destruct signals, that initiate the turn on the detonator firing circuit, occur at Pin A and at the PTB clock count specified on Dwg. 9332388 (see 3.6.8). After 64 min., 128., max., additional PTB clock counts verify that the detonator firing signal (Pin A) turns off and the PA firing signal (Pin B) turns on. This test is to be conducted before the SD1 self-destruct test (4.5.5.9). (Non-destructive test).

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4.5.5.9 SD1 self-destruct signal test. With the electronic assembly under test connected to the test fixture, verify that the SD1 self-destruct signals, that initiate the turn on of the detonator firing circuit, occur at Pin A and at the PTB clock count specified on Dwg. 9332388 (see 3.6.7). After 64 min., 128 max., additional PTB clock counts verify that the detonator firing signal (Pin A) turns off and the PA firing signal (Pin B) turns on. (Non-destructive test).

4.5.5.10 Target sensor functioning test. The potted electronic assembly under test connected to the test fixture shall be tested at the magnetic latitude extremes specified on Dwg. 9332388 for the ambient all-fire and no-fire levels the firing angle specified on the drawing. High level cut-off functioning and spurious outputs shall also be checked at this assembly level. Observation shall be made for compliance with the applicable requirements of 3.6.10. (Non-destructive test).

4.5.5.11 PTB oscillator circuit check. With the electronic assembly under test connected to the test fixture and both the PTB and TTB oscillator circuits operating at their natural frequencies and after electronic arming has occurred, verify that the oscillator check circuit is operational by confirming that the oscillator check circuit initiates the turn on of the detonator firing circuit within 2 TTB clock counts after the PTB oscillator is stopped. Observation shall be made for compliance with 3.6.11. (Non-destructive test).

4.5.5.12 Firing circuit energy. With the electronic assembly under test connected to the test fixture, verify that the piston actuator, and detonator firing circuits meet the requirements specified in 3.6.12, 3.6.12.1, and 3.6.12.2. (Non-destructive test).

4.5.5.13 Low temperature test. With the electronic assembly under test at low test temperature of minus 40° -5°F and connected to the test fixture, verify that the electronic assembly complies with the requirements of paragraph 3.7. The electronic assembly is to be conditioned at minus 40° -5°F for two (2) hours (minimum) prior to the start, and maintained for the duration of this test. (Non-destructive test).

4.5.5.14 High temperature test. With the electronic assembly under test at the high test temperature of 145° + 5°F and connected to the test fixture, verify that the electronic assembly complies with the requirements of paragraph 3.8. The electronic assembly is to be conditioned at 145° + 5°F for two (2) hours (minimum) prior to the start, and maintained for the duration of this test. (Non-destructive test).

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4.5.6 Soldering procedures. All soldering procedures with respect to pre and post cleaning, touchup or rework, and workmanship (as applicable to the requirements of para. 3.9) shall be submitted to the government for approval (see 6.7).

4.5.6.1 Printed wiring board cleaning procedures. Printed wiring board assembly cleaning procedures shall be documented supplied to the government in accordance with Data Item DI-T-5372 (see 6.7).

4.5.7 Safety determination and circuit integrity verification prior to assembly of test mine assembly.

4.5.7.1 Verification of safe condition of S&A assembly. Prior to assembly to the Electronics, the safe position of the S&A slider assembly shall be verified. A flush pin gage shall be used to measure safe position of slider assembly. S&A inspection ports shall be used to verify presence and safe postion of the other safety components. Observation shall be made for compliance with requirements of 3.10. Any S&A failing the requirement shall be classed defective and not used for test. (Non-destructive test).

4.5.7.2 Electronics. Prior to assembly of the Electronics - MCD Assembly to the S&A Assembly, resistance/continuity and battery voltage checks of each assembly shall be made to assure safe condition of the battery and positive shorting of the firing capacitor, and the integrity of electrical circuits listed in Table II. A ten (10) megohm input impedance voltmeter shall be used to test battery voltage and a milliohmmeter having a test current of one milliamp or less shall be used to perform electrical checks listed in Table II. Observation shall be made for compliance with requirement of 3.11. Any assembly failing any requirement shall be classed defective and removed from the lot. (Non-destructive test).

TABLE II

ELECTRICAL VERIFICATION CHECKS OF ELECTRONICS-MCD ASSEMBLY

	Electrical Check	Test Points	Requirements
1.	Shorting Bar Across Firing Capacitor	6 to 9	.2 to .7 ohms
2.	Unactivated Battery	8 to 3 3 to 9	0.005 volts max.
3.	Fuse and I.D. Check	12 to 9	20K ± 5% ohms
4.	Fuse and I.D. check	11 to 9	20K ± 5% ohms
5.	Shorting bar in parallel with EBP plus MCD	G to F	.19 to .63 ohms
6.	MCD in parallel with EBP plus shorting Bar	D to F	.85 to 1.4 ohms

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4.5.8 Material contrc . The contractor shall submit to the technical agency listed in 6.5, a material control document in accordance with paragraph 3.13. This document shall detail the procedures that will be used to assure that only components, material, and subassemblies from qualified suppliers are used to manufacture the BLU 91/B Electronic-MCD Assembly (See 6.4).

5. PACKAGING

5.1 Packaging. The Electronic-MCD assembly shall be preserved and packaged in accordance with MIL-E-17555 with the following mandatory reuirements: The terminal pads shall be wrapped with conductive foam material and the items shall be packaged in a fiberboard box and separated by loose partitions and separators as required. Cushioning as needed shall be used; the fiberboard box shall be sealed in a barrier bag conforming to Type I, Class F, Style 1, Spec. MIL-B-117. (Material used to fabricate the barrier bag must include metal foil).

5.2 Packing. The Electronic-MCD assemblies in box and barrier bag shall be packed in accordance with Level B of MIL-P-14232 and Code of Federal Regulations, Title 49 with the following mandatory requirements: The outer packing box shall be wood conforming to Class 1, Style 4, of PPP-B-621 or fiberboard conforming to Spec. 12H, 23F, or 23H of code of Federal Regulation, Title 49. Sufficient cushioning and separators must be used between the barrier bag and outer box to prevent damage to the items or tearing of the barrier bag.

5.3 Marking. The marking of the outer box shall be in accordance with MIL-STD-129, and DOT marking per Code of Federal Regulation, Title 49. Marking indicating the components are fragile electronics shall be employed.

5.4 Shipping. When Electronic-MCD assemblies from more than one lot are shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing the lots in transit.

6. NOTES

6.1 Intended use. The Electronic-MCD Assemblies covered by the specification are intended for use in Mine, Antitank: BLU 91/B.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit equipment designs as required to: Commander, U.S. Army Armament Research and Development Command (ARRADCOM), ATTN: DRDAR-QAT-I, Dover, New Jersey 07801.

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6.4 Submission of plans for approval. The plans required in Section 4.5.4.4 and 4.5.8 shall be submitted for approval to Commander: ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAT-M.

6.5 Submission of process control documents. Process control documents as specified in 3.12 shall be submitted in accordance with DI-P-1604 (Mod) with attachment 1, to: Commander, ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAT-M.

6.6 Distribution of ammunition data cards, inspection and test results, process controls. Distribution of ammunition data cards, inspection, and test results shall be sent to Commander: ARRADCOM, Dover, NJ 07801, ATTN: DRDAR-QAT-M.

6.7 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, contract Data Requirments List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporaed into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement	Applicable DID No.
6.5	Manufacturing Methods Report	DI-P-1604 (Mod)
4.5.6.1	Printed wiring board cleaning procedures	* DI-T-5372 (Tailored)
4.4.2.1	Quality Inspection Defect Report (Critical Defects Only)	DI-R-1721
	Quality Inspection Test Demonstration and Evaluation Report	** DI-R-1724 (Tailored)

* Delete a, b, c, d of previous Paragraph 1, Block 10, DD Form 1664, and substitute contractor designated parameters and limits.

** DI-R-1724 (Tailored) with Block 10 modifications dated 15 October 1981.

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

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6.8 First article. Provisions for first article as specified in Sections 3 and 4.

6.9 Definitions.

a. Premature Arming is defined as arming of the S&A Assy (actuation of micropiston actuator) prior to minimum arming time specified on Dwg. 9332388.

b. Premature Detonation is defined as detonation of an armed S&A Assy. prior to the minimum electronic arm time specified on Dwg. 9332388.

c. Early Self-Destruct is defined as detonation of an armed S&A occurring at or after the minimum electronic arm time specified on Dwg. 9332388 but prior to the set SD time period, or prior to being functioned by sensor mode.

d. Low Order Detonation for the Test Mine Assembly is defined as detonation of an armed S&A where only two main charge leads detonate properly.

e. Hazardous Dud is defined as an armed test mine assy. which detonates after the programmed SD time has elapsed by any of the following:

(1) Actuation by prime sensor.
(2) Being disturbed.
(3) Power source run-down.

f. Non-Hazardous Dud is defined as a test mine assy. which will not detonate by conditions 1, 2, or 3 listed under Hazardous Dud.

g. Sensor failure is defined as a test mine assy. which will not detonate when subjected to the test signal of Dwg. 9332388.

h. Transportation Vibration failure is defined as a test mine assy. that does not meet the requirements of 3.4.1, or has the borerider function as a result of vibration.

i. Impact failure is an Electronic-MCD Assy. that is non-functional due to damage sustained during impact, or the MCD assy. moves with respect to the electronic assembly.

6.10 Test mine assemblies. To prevent damage to the Electronic-MCD Assembly, the contractor may, with prior approval from the Government, cut or remove the MDF assembly from the electronic side of the S&A.

6.11 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock

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Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARRADCOM.

Custodian:
Army AR

Preparing Activity
Army AR

(Project 1345-A173)

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APPENDIX

SUPPLEMENTAL SOLDERING PROVISIONS

10 SCOPE

10.1 Scope. The appendix details supplemental soldering provisions to MIL-S-45743 for hand soldering and to MIL-S-46844 for machine soldering. The appendix is not a mandatory part of the specification. This information contained herein is intended for information only.

20 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30 REQUIREMENTS

30.1 Exception/clarification to requirements of MIL-S-45743. The following exceptions/clarifications to reference MIL-S-45743 paragraph titles shall apply.

a. Control of airborne contaminants. Delete the sentence, "Positive pressure shall ... between them."

b. Soldering irons and resistance soldering irons. Delete the sentence, "The soldering iron shall ... or standby condition.", and substitute the sentence, "The soldering iron shall be a soldering iron approved by the technical agency or a soldering iron that attains a temperature no more than 10°F (5.6°C) above the control temperature when in an idling or standby condition."

c. Printed board terminations., Subparagraph 'b'. Change to read "... is permissible. Leads shall not extend, beyond the terminal pad or printed wiring pattern, far enough to cause less than .015 inches of clearance between conductors."

d. Printed board terminations., Subparagraph 'd'. Change to read, "When leads cannot be clinched per above, they shall extend through the printed board, be visible and properly soldered."

e. Ribbon lead geometry. Add new sentence, "Ribbon lead termination from battery and lead assembly, shall comply with "Printed board terminations", subparagraph 'b'."

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30.2 Exceptions/clarifications to requirements of MIL-S-46844. The following exceptions/clarifications to reference to MIL-S-46844 paragraph titles shall apply.

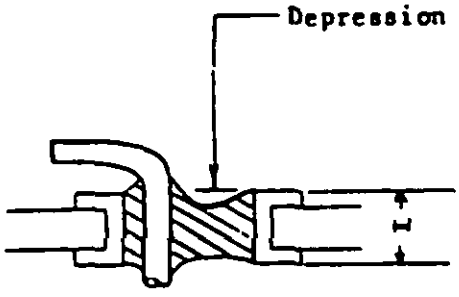
a. Flux application. Change to read "... even coated surface. Any thinner used shall be a product from the flux manufacturer recommended for the specified flux being utilized or a substitute approved by the technical agency."

b. The solder contact. Add: "Bonded components or clinched leads are acceptable restraints provided they are implemented in a manner to assure that the leads do not more during the solder cycle."

30.3 Supplemental solder workmanship standards.

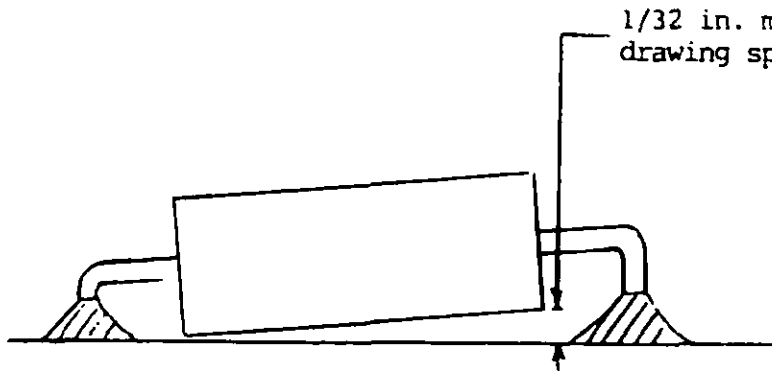
30.3.1 General. The following solder workmanship standards supplement the workmanship requirements contained in MIL-S-45743 and MIL-S-46844 and are provided as additional aids in establishing the minimum acceptance standards for soldering. All examinations will be performed using a minimum 5 diopter lens. Suspected connections may be examined using 10x magnification to determine acceptability.

30.3.2 Lead, solder side. Exposed end (except copper) are visible at the end of the lead. Lead is not closer than .015" to adjoining conductive circuitry. Maximum solder is permissible. Solder fillet is depressed, but the depression is less than 25% of the board thickness.



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30.3.3 Component alignment.

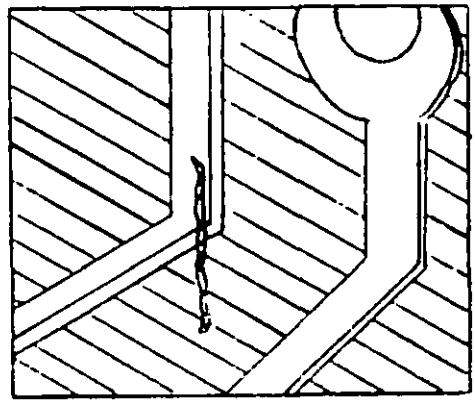


Vertical misalignment of component and printed wiring board is not in excess of 1/32 in. (or as specified on engineering drawing).

30.3.4 Scratches.

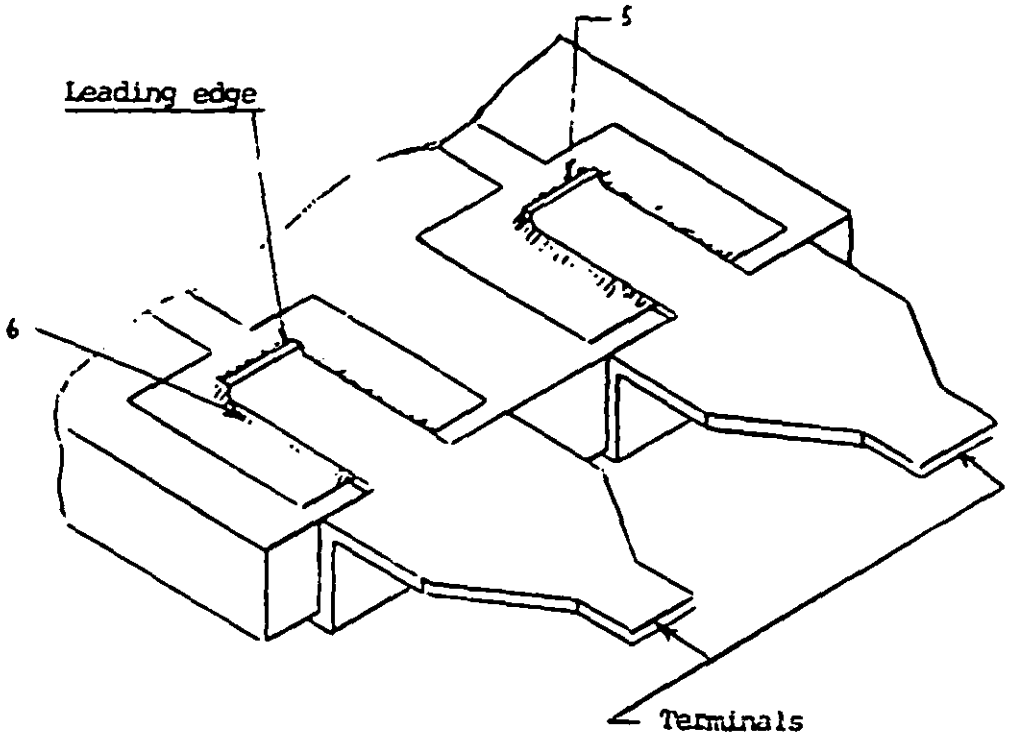
1. Scratch does not reduce cross-sectional area of conductor more than 23%

2. Does not visibly expose base metal.



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30.3.5 Shorting bar soldering. This section described the acceptance criteria for solder connections of terminal to pad on ceramic substrate. All examinations shall be performed with 10x magnification (Microscope).



PREFERRED

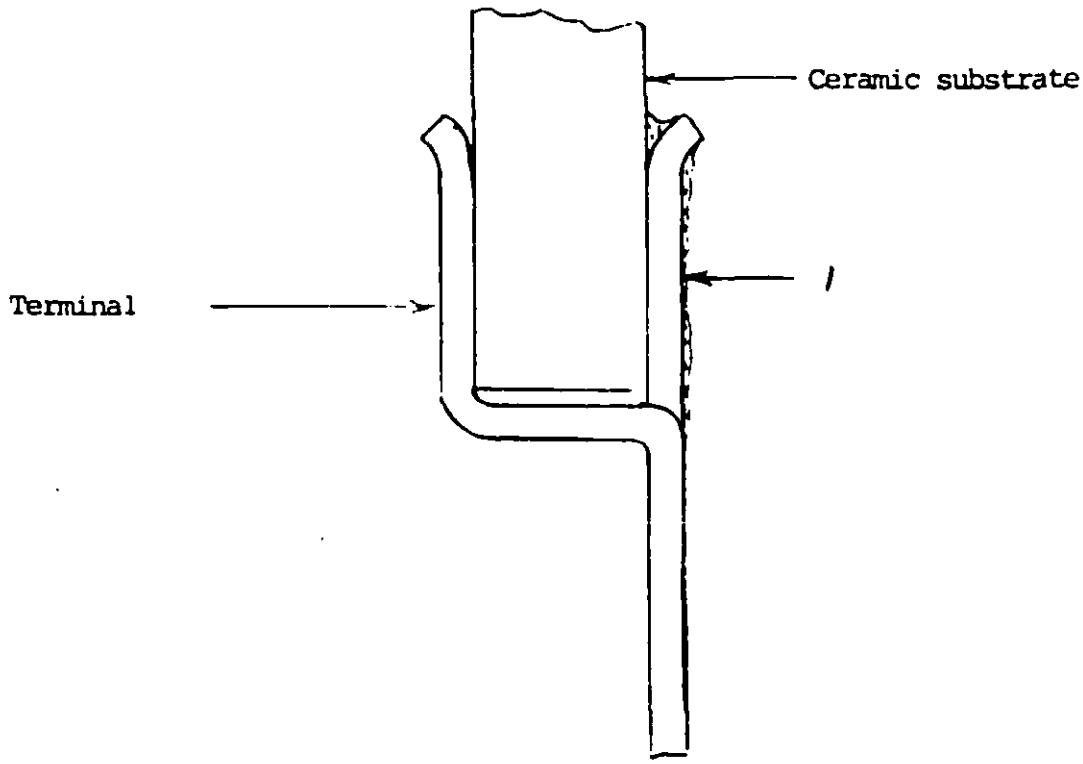
1. Terminals are straight, parallel to and centered on solder pad.
2. Solder fillets are continuous on all edges for the entire length of terminal/pad interface without voids.
3. Solder wetting is evident on terminal edges and solder pad.

ACCEPTABLE

4. Solder is continuous and wetted on the entire leading edge and along both sides for 50% of the terminal/pad interface. No cracks or voids may appear between terminal and solder pad in the solder area.
5. Solder is continuous and wetted on the leading edge and one (1) of two (2) edges of terminal for the entire length of the solder pad/terminal interface. No cracks or voids may appear between terminal and solder pad in solder area. There must be evidence of some solder wetting on opposite edge.
6. Solder fillet, 3/4 thickness of terminal, minimum.

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APPENDIX

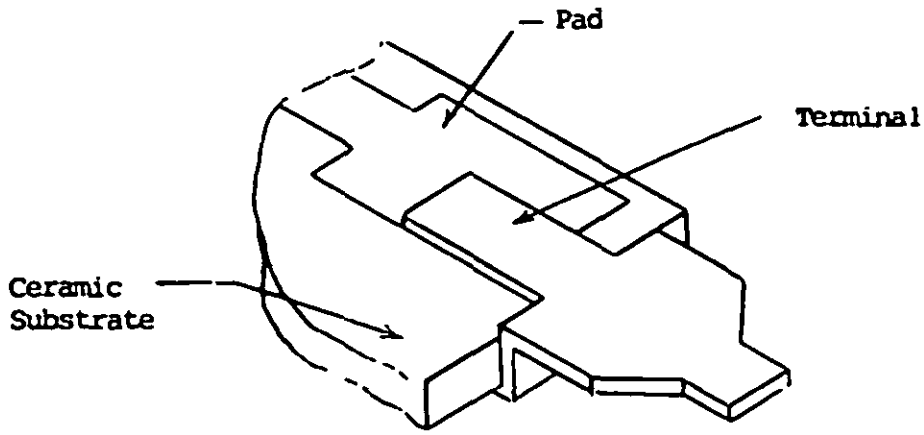


ACCEPTABLE

1. Solder buildup is not greater then one (1) terminal thickness.

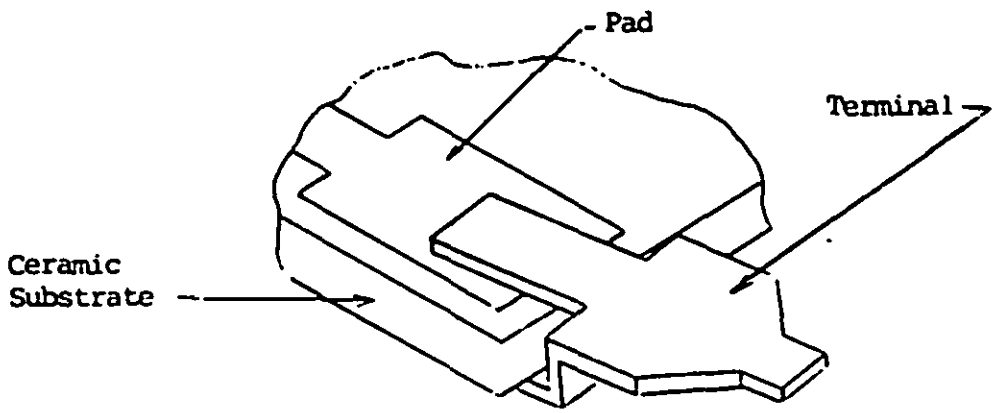
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APPENDIX



ACCEPTABLE MINIMUM

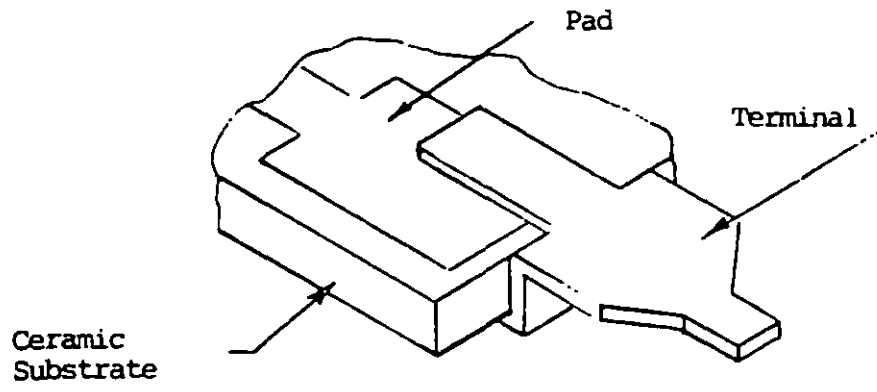
1. Terminal edge coincident with pad edge



ACCEPTABLE MINIMUM

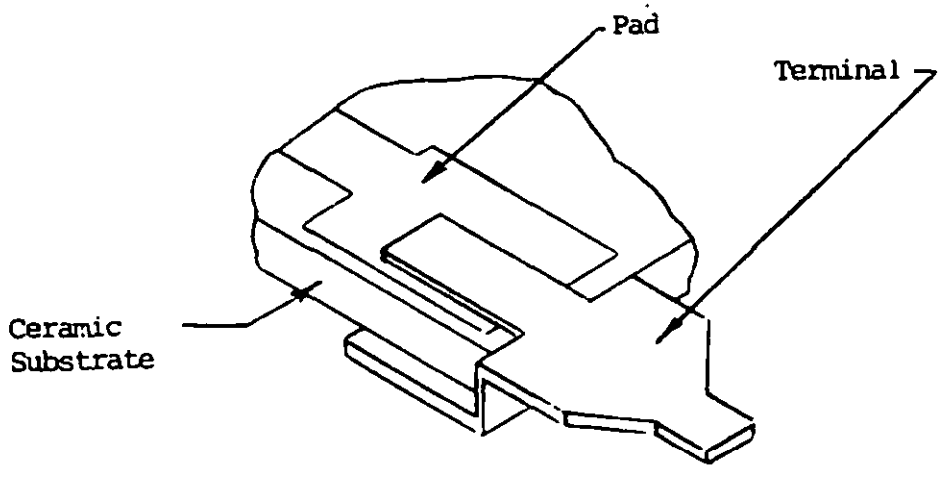
1. Terminal slightly skewed, but does not extend beyond pad edge.

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UNACCEPTABLE

1. Terminal overlaps pad

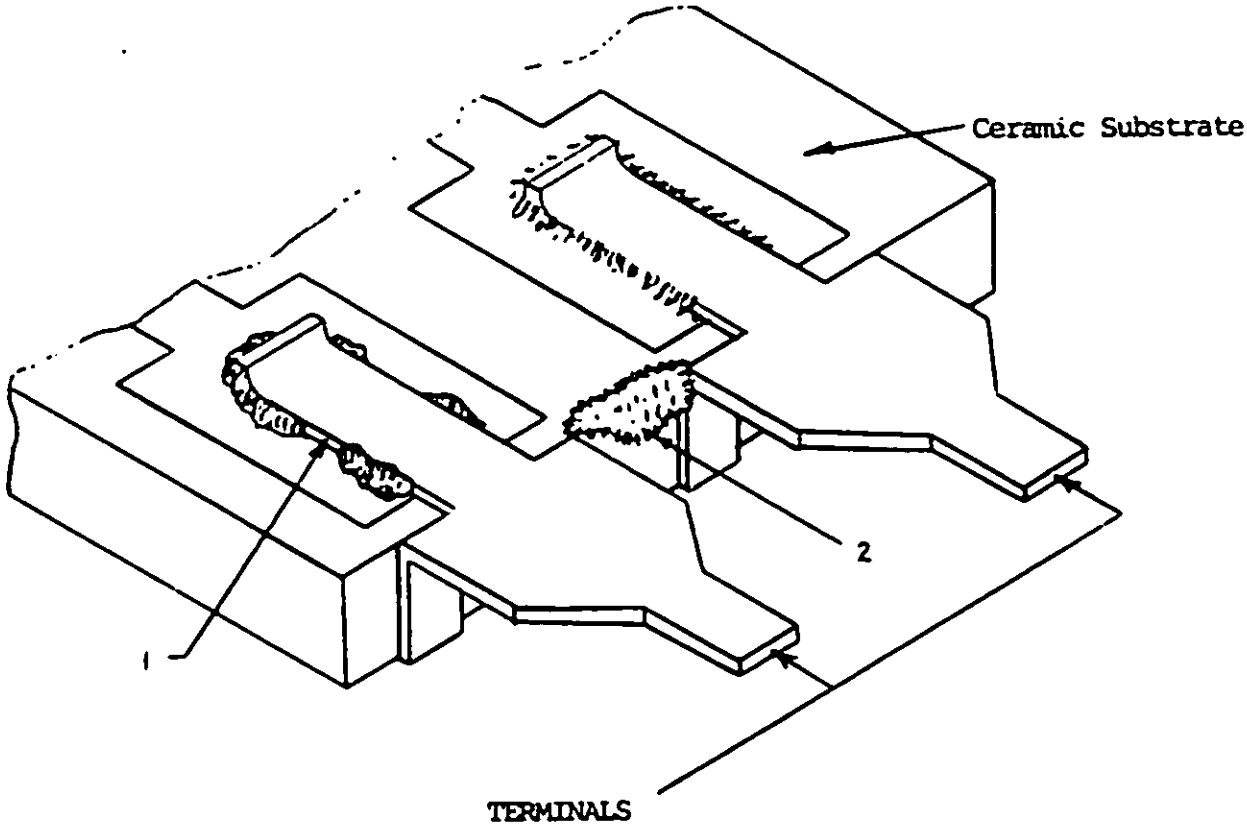


UNACCEPTABLE

1. Terminal overlaps ceramic substrate

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APPENDIX



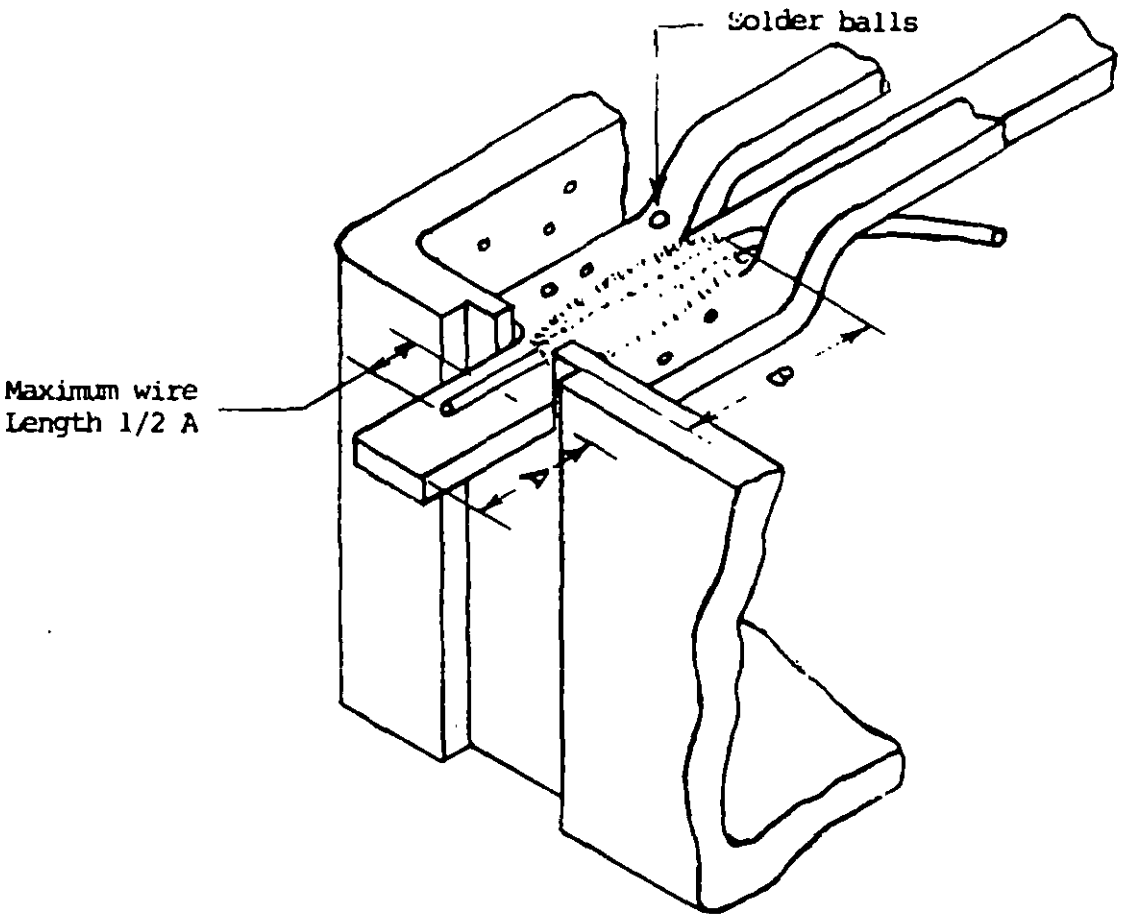
UNACCEPTABLE

1. Gaps in solder fillet exceed Note 4 of figure showing perferred assembly.

2. Bridging between terminals.

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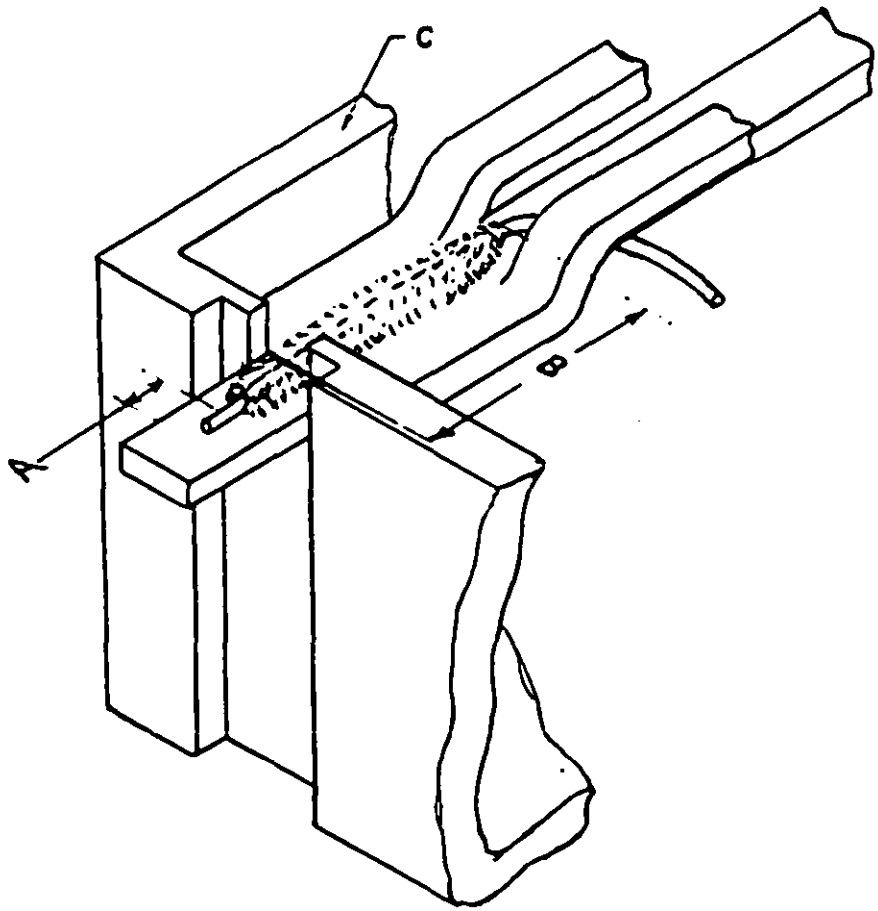
30.3.6 Magnetometer soldering. This section describes the acceptance criteria for solder connections of magnet wire to terminal. All examinations shall be performed with 10x magnification (microscope).



PREFERRED

1. Good wetting of wire and terminal over entire length "B".
2. Solder is free of holes.
3. Contour of the wire visible.
4. Wire is extended along the length of terminal.
5. Protrusion of wire is less than one-half of Terminal length (A).
6. Solder balls are permissible in the potting area.
7. Bare copper end of wire is permissible.
8. Wire routed between terminal tabs, but no mechanical wrap is required.
9. Melted plastic may be present in the vicinity of solder joint (not allowed in solder joint).

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ACCEPTABLE PROTRUSION OF WIRE

1. Protrusion of the wire is above the surface of the terminal, but below the mold surface "C".

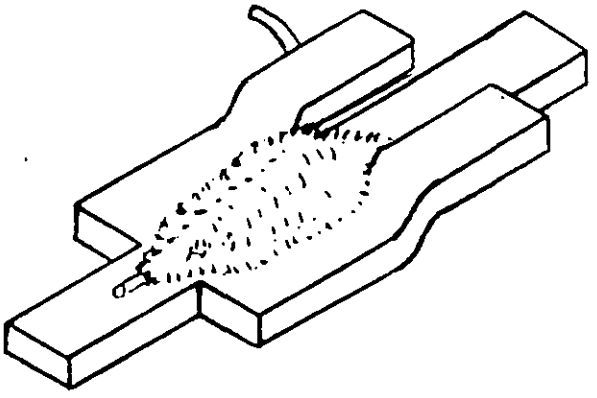
2. The lead must extend into area "A".

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ACCEPTABLE

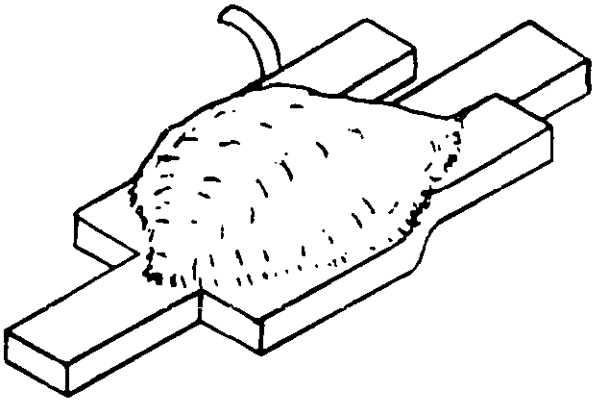
1. Contour of wire not visible but wire ends are visible and show evidence of wetting, entrance and exit.

2. Irregular surface; slightly convex fillet.



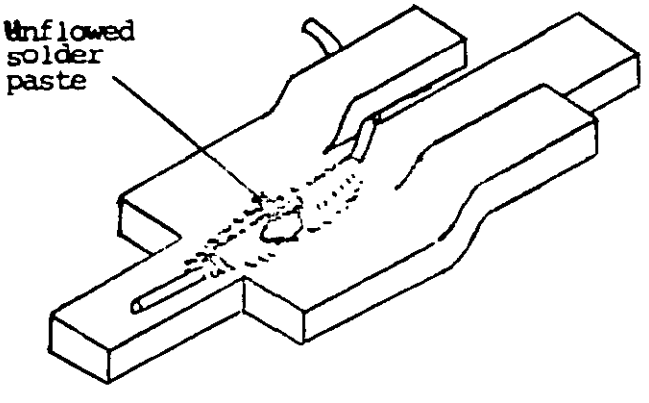
UNACCEPTABLE

1. The end of the wire is not visible



UNACCEPTABLE

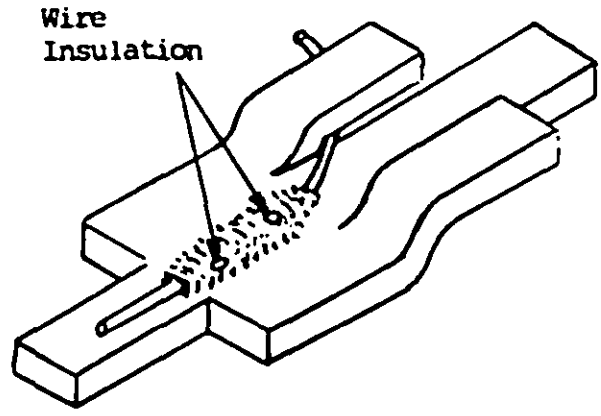
1. Evidence of solder paste in solder connection.



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UNACCEPTABLE

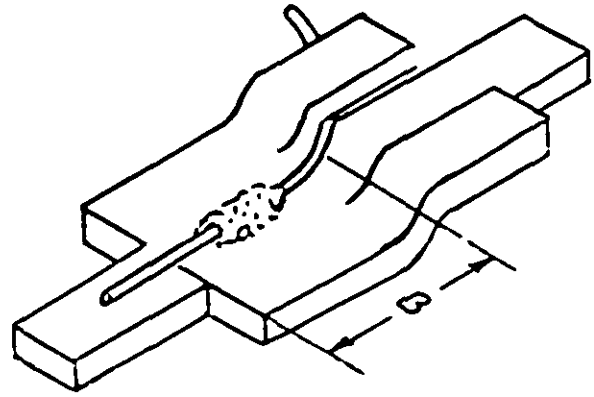
1. Evidence of contamination in solder connection.



UNACCEPTABLE

1. Solder covers less than 50% of length "B".

2. Solder is not continuously wetted.



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APPENDIX

40 QUALITY ASSURANCE PROVISIONS

40.1 Exceptions/clarifications to quality assurance provisions of MIL-S-45743. The following exceptions/clarifications to reference MIL-S-45743 paragraph titles shall apply.

a. One hundred percent inspection. Delete and substitute the following: "Level of inspection. Inspection of soldered connections and associated wiring and leads shall be performed to the level specified in the classification of defects in the product specification. Table II of MIL-S-45743 provides a list of defects with reference to the applicable requirements."

b. Visual inspection. Change to read "... a clear incandescent or fluorescent light should be used for detection of uncovered copper."

40.2 Exceptions/clarifications to quality assurance provisions of MIL-S-46844. The following exceptions/clarifications to reference MIL-S-46844 paragraph titles shall apply.

a. One hundred percent inspection. Delete and substitute the following: "Level of inspection. Inspection of soldered connections and associated wiring and leads shall be performed to the level specified in the classification of defects in the product specification. Table II of MIL-S-45743 provides a list of defects with reference to the applicable requirements."

b. Visual inspection. Change to read "... a clear incandescent or fluorescent light should be used for detection of uncovered copper."

40.3 Supplemental quality assurance provisions.

40.3.1 General. The following quality assurance provisions supplement the provisions contained in MIL-S-45743 and MIL-S-46844 and are provided as additional aids in establishing the minimum acceptance standards for soldering. All examinations will be performed using a minimum 5 diopter lens. Suspect connections may be examined using 10x magnifications to determine acceptability.

40.3.1.1 Major defect inspection.

A board will be defined as defective during inspection if:

(1) The bottom contains any joints defined as unacceptable by MIL-S-45743, MIL-S-46844, or sections 30 and 40 of this appendix.

(2) The top contains any joint defined as unacceptable by the documentation listed in 1 above except twelve (12) joints are allowed to be dewetted. A dewetted joint on the top must have evidence of solder in the hole when viewed from the top.

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40.3.1.2 Minor defect inspection.

a. Inspection shall be on the sample size inspected for major defects.

b. A board will be defined as defective if it contains three (3) or more minor defects.

c. Accept/reject criteria for the lot shall be to a 4.0% AQL, MIL-STD-105, Level II, normal sampling plan.

d. Minor defects are defined as the following:

(1) Conductive surfaces closer than .010" but greater than .005" assembly characteristics only, not relative to the printed wire board construction.

(2) Copper showing.

(a) Clipped lead - evidence of good wetting and reflow must be present.

(b) Non-clipped lead - characteristic of good solder joint, i.e., wetting, etc., must be present.

(3) Solder splash (attached and not shorted).

(4) Solder balls less than .005".

50 PACKAGING

This section is not applicable to this appendix.

60 NOTES

60.1 Definitions.

60.1.1 Voids. All holes, pits, and voids regardless of size are defined as voids.

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